Reconnaissance paleomagnetic study of igneous rocks from the eastern sector of the Labrador Trough[1]

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A paleomagnetic study has been carried out on 75 samples of metagabbro and metaperidotite from 14 sites in the Labrador Trough. The sites were distributed over a distance of 500 km and are situated in sills intruded into formation near the top of the trough stratigraphic column. Four components (A, B, C, and D) were obtained after alternating field (AF) and thermal demagnetization. The mean directions of magnetization are A, 143°, +33°, α_{95} = 14.4°, k = 15.4, 8 sites; B, 69°, +51°, α_{95} = 10.3°, k = 22.9, 10 sites; C, 278°, +15°, α_{95} = 7.7°, k = 28, 14 sites; and D, 200°, +16°, α_{95} = 13.3°, k = 16.2, 9 sites. All components are secondary. The fold test is negative, and consequently no primary magnetization has been preserved in the isolated components. Component C may have erased all others in zones of highest metamorphism. The paleopoles obtained for components D, A, and C are, respectively, 270°E, 23°S (δ_p = 10°, δ_m = 18°); 327°E, 9°S (δ_p = 11°, δ_m = 20°); and 19°E, 11°S (antipole) (δ_p = 10°, δ_m = 17°). The paleopole positions obtained for these three components are compared with the existing late Aphebian and early Paleohelikian apparent polar wander (APW) path for North America and fall in the 1750–1850 Ma interval.

Une étude paléomagnétique a été complétée en utilisant 75 échantillons de métagabbro et de métapéridotite prélevés sur 14 emplacements dans la zone de plissements du Labrador. Les emplacements d'échantillonnage sont étalés sur une distance de 500 km, dans des filons-couches pénétrant les formations près du toit de la colonne stratigraphique de la zone de plissements du Labrador. La désaimantation par champs alternatifs et par traitements thermiques a fourni quatre composantes (A, B, C et D). Les directions dominantes d'aimantation sont A, 143°, +33°, α_{95} = 14,4°, k = 15,4, 8 emplacements; B, 69°, +51°, α_{95} = 10,3°, k = 22,9, 10 emplacements; C, 278°, +15°, α_{95} = 7,7°, k = 28, 14 emplacements; et D, 200°, +16°, α_{95} = 13,3°, k = 16,2, 9 emplacements. Toutes les composantes sont secondaires. Le test de plissement est négatif et par conséquent l'aimantation primaire des composantes isolées est disparue. La composante C a pu effacer toutes les autres dans les secteurs où le métamorphisme fut le plus intense. Les paléopôles obtenus pour les composantes D, A et C sont, respectivement, 270°E, 23°S (δ_p = 10°, δ_m = 18°), 327°E, 9°S, (δ_p = 11°, δ_m = 20°) et 19°E, 11°S (antipôle) (δ_p = 10°, δ_m = 17°). Les positions des paléopôles obtenues par ces trois composantes sont portées sur la courbe de la dérive apparente des pôles de l'Amérique du Nord pour l'Aphébien supérieur et le Paléohélikien inférieur, et elles se situent dans l'intervalle 1750–1850 Ma.

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Introduction

Aphebian paleomagnetic results obtained from the Circum-Ungava belt are extremely difficult to interpret. The present paleomagnetic study increases the data set and is an attempt to better understand the mode and time of formation of the Labrador Trough in relation to the Circum-Ungava Belt. If a component of magnetization is primary, the paleomagnetic results can be compared with those of other segments of the Circum-Ungava belt, e.g., from Cape Smith (Fujiwara and Schwarz 1975), from Hudson Bay (Schwarz 1976; Schwarz and Fujiwara 1981), from Proterozoic outliers in northern Quebec (Seguin *et al.* 1981*b*; Schwarz and Freda 1983), and from areas close to the Grenville Front (Fahrig and Chown 1973; Fahrig *et al.* 1974). If only secondary components are isolated (e.g., Park 1977), they may reveal the thermal and (or) dynamometamorphic history of the trough and its evolution as part of the Circum-Ungava Belt.

Geological setting

The Labrador Trough, the erosional remnant of an Aphebian geosyncline, consists largely of sedimentary rocks in the West and basic igneous rocks (basalts and gabbros) in the east. These rocks form two longitudinal zones, termed the miogeosyncline and the eugeosyncline, respectively. The stratigraphy of the trough has been outlined by Dimroth (1978), Dimroth *et al.* (1970), and Wardle and Bailey (1981). The rocks of the trough experienced one major period of metamorphism and deformation, the Hudsonian Orogeny, which closed at about 1750 Ma (Stockwell 1982). This deformation was very complex and resulted in interfering fold patterns (Dimroth 1970); two or more cleavages are commonly observed (Clark 1979).

Fourteen sites in the Labrador Trough were sampled; these extended from near Baie aux Feuilles in the north to Doublet Lake in the south, a distance of nearly 500 km. For site locations, see Fig. 1.

All sites are underlain by intrusive rocks of the eugeosyncline (Fig. 1). The rocks at eight sites (1980-3, 1980-5, 1980-6, M1, M3, M11, M12, M13) consist of ophitic and subophitic metagabbro; those at one site (1980-1) consist of glomeroporphyritic metagabbro, those at three sites (1980-2, M4, M5) of quartz-bearing metagabbro, those at one site (M2) of serpentinite (altered dunite), and those at one site (1980-4) of granitoid material (probably a siliceous phase of a differentiated gabbro sill).

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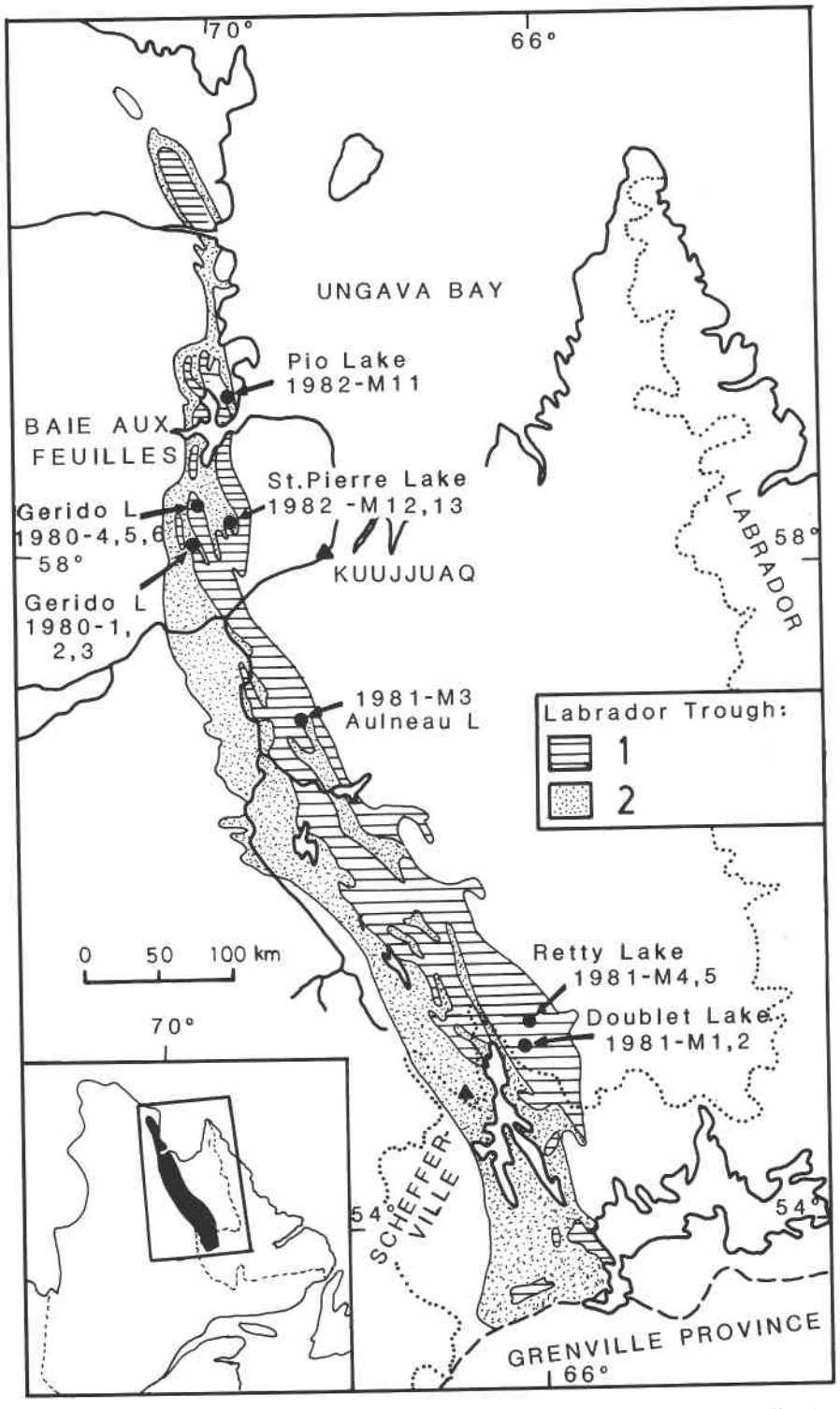


FIG. 1. Site location map showing name, sample year, and number of each site. Legend: Unit 1 (horizontal lines) — igneous rocks (gabbro, peridotite, basalt) and intercalated sedimentary rocks. Units 2 (stippled) — sedimentary rocks (sandstone, mudrock, dolomite, iron formation, etc.).

The igneous rocks at all sites were intruded late in the depositional history of the trough and hence have roughly the same age. The enclosing sedimentary and effusive rocks belong to the third depositional cycle, the Doublet Group (Baragar 1967; Dimroth 1970). The basic sills were therefore emplaced before onset of the Hudsonian Orogeny.

The rocks at all sites were strongly recrystallized during the Hudsonian Orogeny. The highest metamorphic grade in the region, the lower amphibolite facies, was reached at Pio Lake in the northern sector. At St. Pierre Lake in the northern sector the biotite zone of the greenschist facies was reached, whereas at Gerido Lake, also of the northern sector, and at all other sites the chlorite zone of the greenschist facies was attained.

The samples of lower amphibolite facies consists of plagioclase, actinolite, epidote, and sphene; they are devoid of chlorite. Gabbros metamorphosed to the greenschist facies con-

TABLE 1. Relations among components, metamorphism, and mineralogy

Site	Locality	Regional metamorphism (facies)	Components	Amphibole III*	Po	Ru	Il	Mt
M-11	Pio Lake	Amphibolite	C	×	×			
M-12	St. Pierre Lake	Greenschist (biotite zone)	C	×			×	
M-13	St. Pierre Lake	Greenschist (biotite zone)	C	×				
M-3	Aulneau Lake	Greenschist (chlorite zone)	B C D	×	×			
M-4	Retty Lake	Greenschist (chlorite zone)	B C D	—	×	×		
M-5			A B C D	—	×	×		
M-1	Doublet Lake	Greenschist (chlorite zone)	A C D	×	×	×		
M-2			B C D	—			×	
1980-4	Gerido Lake	Greenschist (chlorite zone)	A B C D	—		×		
1980-5		Greenschist (chlorite zone)	A B C	×		×		
1980-6		Greenschist (chlorite zone)	A B C D	×	×			
1980-1		Greenschist (chlorite zone)	A B C D	—		×		
1980-2		Greenschist (chlorite zone)	A B C D	—			×	
1980-3		Greenschist (chlorite zone)	A B C	—	×	×	×	

NOTES: Po = pyrrhotite; Ru = rutile; Il = ilmenite; Mt = magnetite; × = present.
*Amphibole III indicates advanced state of replacement of primary mafic minerals (Dimroth and Ressler 1978); × indicates abundant amphibole III; otherwise, it is present but not abundant.

sist of albite, epidote, actinolite, chlorite, sphene, + biotite, + quartz, + rutile, + carbonate. Samples from all sites contain clusters of fine-grained sphene (leucoxene) that, in the chlorite zone, commonly have cores of fine-grained rutile and (or) ilmenite. Traces of pyrrhotite and (or) chalcopyrite occur in most of the metagabbros. Magnetite was not identified microscopically in the samples of metagabbro (Table 1). The magnetic characteristics and microprobe analyses of these rocks, on the other hand, suggest that the rock magnetization is carried by titanomagnetite and a trace of magnetite (Table 1). It would appear, therefore, that these minerals are present at submicroscopic scale. In large part, the primary oxide minerals appear to have been converted to sphene containing a little rutile and (or) ilmenite.

Paleomagnetic results

Field sampling and labratory treatment

Hand samples were oriented in the field with a magnetic compass and subsequently cored in the laboratory. Each paleomagnetic site typically comprises five or six independently oriented samples distributed over several metres of outcrop. Rock units with marked fabric, indications of anomalous metamorphism, or a medium to high degree of alteration were avoided.

Natural remanent magnetizations (NRM) and ferrimagnetic susceptibilities were measured according to the procedure described by Seguin *et al.* (1981*a*) (Fig. 2). The magnetization was measured with a digital Schonstedt spinner magnetometer (DSM-1), and the volume susceptibility was measured at room temperature in the Earth's field ($H = 0.05$ mT), using a Cenco MS-3B susceptibility bridge. Detailed alternating field (AF) and thermal demagnetizations were then carried out on the bulk of the collection. AF demagnetization was conducted with a three-axis tumbler operating in residual fields of about 5 nT. A Schonstedt thermal demagnetizer (TSD-1) with residual fields of less than 15 nT was also used.

At least one specimen of each sample was progressively demagnetized in steps of 2.5−10 mT up to a maximum field of 100 mT or to fields at which the magnetizations could no longer be accurately measured or directional behaviour became erratic. The performance of the AF demagnetizer built at University of Laval was largely improved by adding three large mu-metal cylinders around the solenoid (Seguin 1975*b*); a three-axis tumbler was used with this unit. Progressive thermal demagnetization was undertaken on each specimen in steps of 100°C from room temperature to 400°C; thereafter, incremented steps were reduced, approaching 25° near the Curie point(s). The response of the magnetic vector to these treatments was often quite different for individual samples from the same site, and consequently detailed progressive cleaning was needed for each sample. Site-mean directions of magnetization were calculated using Fisher's (1953) statistical method. In slowly cooled metamorphic rocks, more than one component of magnetization was often detected in individual samples, and the grouping required to perform this calculation was determined by inspection of the demagnetization plots.

Eleven (11) of the 14 sampling sites have either three or four stable magnetization components (see Table 1). However, no more than two components were isolated in individual samples, and only eight individual specimens contained two components. The single components were defined with stereoplots and (or) Zijderveld's (1967) diagrams. Figure 3 shows examples of specimens having two components of magnetization. In the case of samples with two components, vector subtraction was occasionally used in addition to the two other extraction techniques. Except for a dozen individual specimens, no specific difficulties were encountered in separating the components. There is no substantial difference between the directions obtained from single-component samples and those obtained from two-component samples.

1982 Pio Lake − St. Pierre Lake areas

All the samples from this area were collected from gabbro sills. The range of unblocking temperatures (T_{UB}) is narrower (400−525°C) than for the two sampling areas discussed below, and the median destructive field is located in the 10−30 mT range. Only one component of magnetization (northwest shallow) was detected in these samples. Thermal demagnetization was definitely more efficient (in particular, more conclusive end points) than AF treatment for resolving the magnetization.

1980 Gerido Lake area

Five of the six sites from the Gerido Lake area are in gabbro

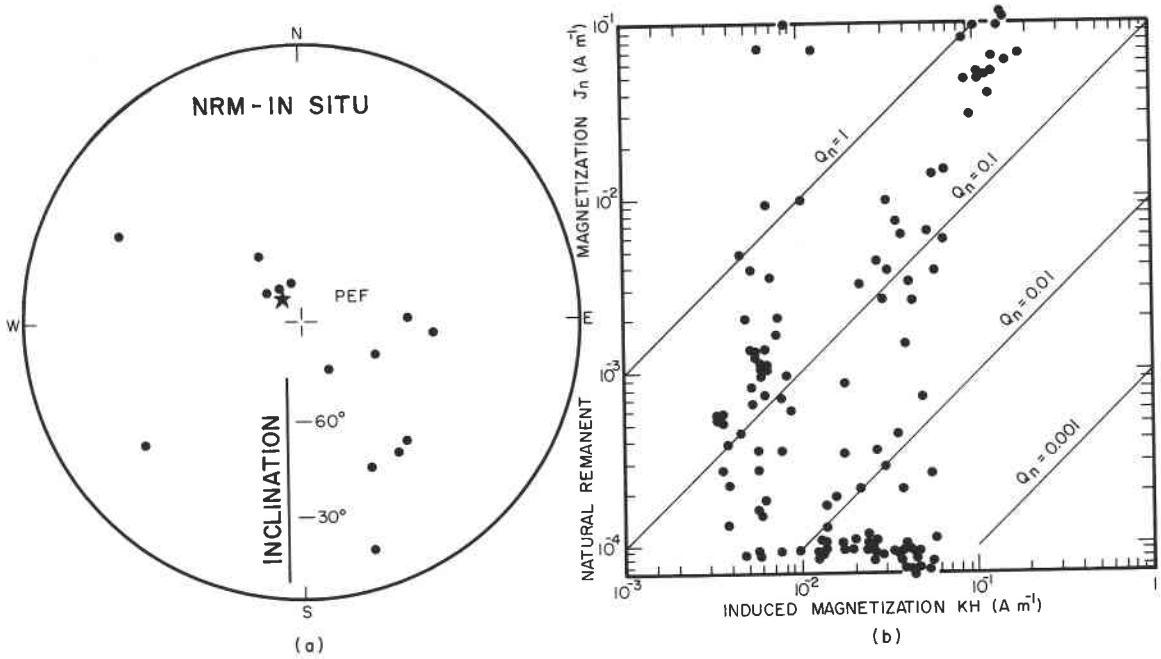


FIG. 2. (*a*) Mean NRM site directions for igneous rocks from the Labrador Trough plotted in a Schmidt net. All directions are down. The star represents the present Earth's field (PEF) (325°, 80°N) at the sampling region. (*b*) Plots of intensity of NRM (J_n) for individual samples against the induced magnetization (KH), with resulting slope lines indicating the values of Koenigsberger's ratio, Q_n.

sills. The samples exhibit a range of unblocking temperature spectra; a few have square-shoulder thermal-decay curves with unblocking temperatures in the range 500–550°C, whereas a larger number of sites exhibit a low unblocking temperature component at 400°C. In general, AF demagnetizations produced a systematic shift of the vector from south-southeast with intermediate or steep positive inclinations to south-southeast or south-southwest or east-northeast (R) with shallow or intermediate inclinations. A few sites exhibit little movement of the vector but substantial drop in intensity with AF treatment. Optimum grouping of site vectors is achieved by AF demagnetization between 15 and 60 mT. The AF and thermal results generally agree.

1981 Doublet Lake – Retty Lake – Aulneau Lake areas

Except for one serpentinized peridotite site (M-2), the samples were collected from gabbro sills. A few samples have square-shoulder thermal-decay curves associated with unblocking temperatures in the range 500–525°C, whereas a somewhat larger number of samples exhibit only a low unblocking temperature component in the 300–400°C range. Most commonly, both low and high unblocking temperature components were detected in the same samples. The western magnetization directions of component C correspond generally to unblocking temperatures in the temperature range of 450–540°C, whereas the south-southwest directions of magnetization of component D are frequently characterized by T_{UB}'s in the 300–450°C range. The T_{UB} ranges are much more variable for component A and B. In general, AF demagnetization produced a rapid movement of the vector in the range 2.5–10 mT. A component was then isolated in the 7.5–20 mT interval. Most commonly, this low field component coincided with the low temperature component. Occasionally a high field component (30–60 mT range) was also isolated. This component may coincide with the high temperature component, but this is not always the case, and it too may correspond to a low temperature component. Frequently, an NRM direction with intermediate to steep inclination decreases to a shallow inclination during AF treatment.

Components of magnetization

Four components of magnetization (A, B, C, and D; see Table 2) were isolated in the areas sampled. The only component isolated in sites from Pio and St. Pierre lakes is component C; its mean direction is west-northwest, intermediate downwards. In the Gerido Lake sector, component A is the most common; its mean direction is south-southeast, intermediate downwards. Components A and C are three times as common as component B. Component D is the least common component. In the areas farther south, component C is more common than components B and D, and component A is uncommon. Figure 4 shows one typical vector diagram of both AF and thermal demagnetization behaviour for the two most frequent components (A and C). Figure 5 shows the means per site and the mean for each magnetic component.

Fold test

The metalavas and metagabbros of the Labrador Trough were severely deformed in the course of the dynamometamorphism of the Hudsonian Orogeny. Even though the strikes of the sampled strata are quite constant and the fold plunges generally shallow (15–20°), the dips are very variable, ranging from intermediate to steep (with some strata overturned). The results of the fold test (Table 2) in which overturned strata have been taken into account are significantly negative (–) on all components (A, B, C, D) at the 95% confidence limit (McElhinny 1964) except for component B of the 1980 Gerido Lake area and possibly of the 1981 Lake Doublet – Lake Retty – Lake Aulneau areas, on which the fold test is negative but not at a significant level for the 95% confidence limit. Conseuently, it appears that all the components of magnetization are syn- or postfolding, and they do not preserve a record of the

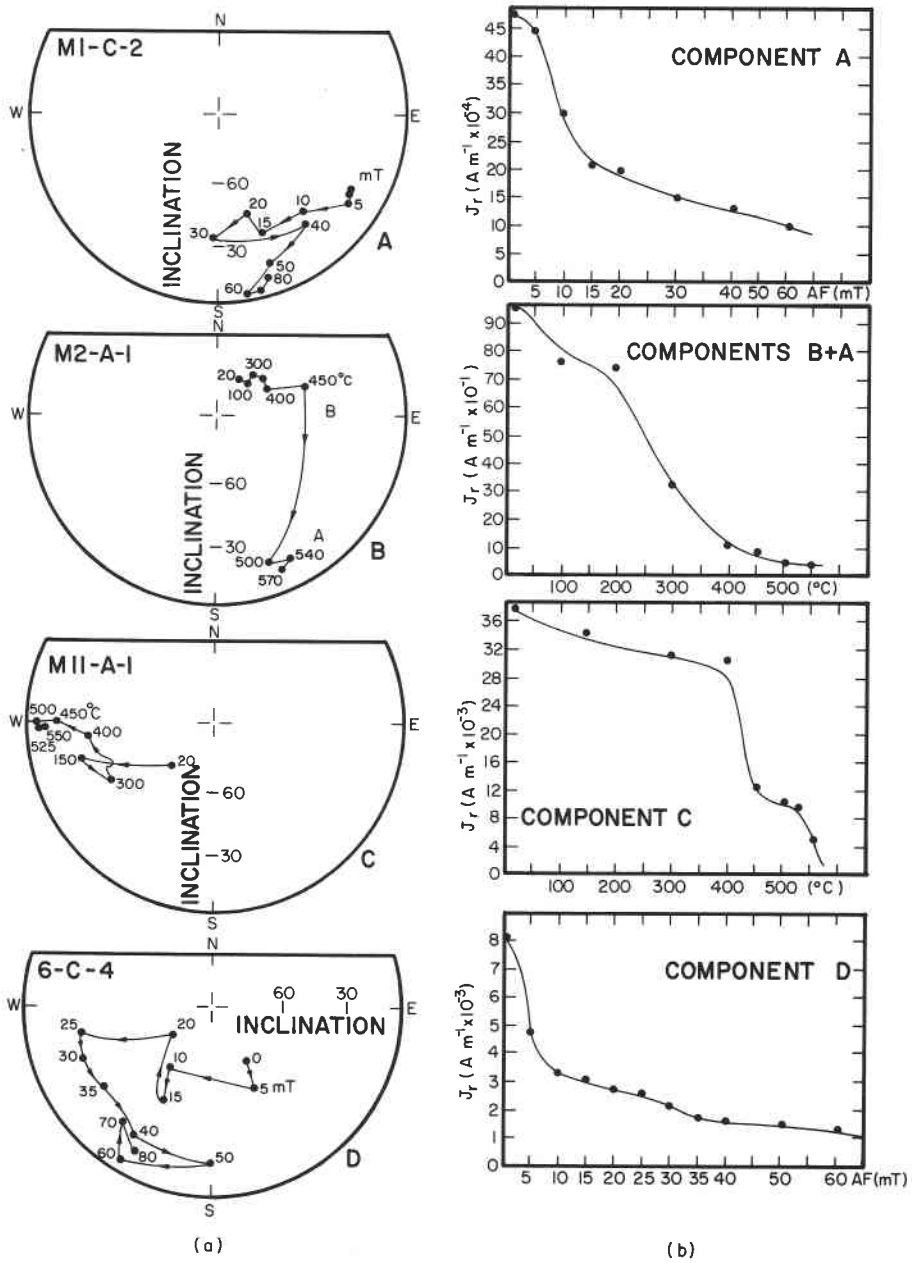


FIG. 3. (*a*) Stereographic plots showing variation of the magnetization directions of four pilot specimens of metagabbros as a function of AF demagnetizing strength or temperature. (*b*) Remanent intensity of the same pilot specimens of metagabbros as a function of the demagnetizing field strength and temperature. A typical example of each of the four components of magnetization is represented.

primary age of magnetization (intrusion).

Discussion of the results

Magnetic record

Within the individual areas, there is no clear correlation between site-mean direction and unblocking temperature, coercive force, or geographic location. All four components were detected in the Gerido Lake area, and in areas to the south components A and C have slightly higher T_{UB} and H_C spectra. In the northern sector of the Labrador Trough, the metamorphic grade increases slowly but progressively going north (Dimroth and Dressler 1978). The northernmost samples from Pio Lake are of lower amphibolite grade, and the samples from Gerido Lake in the chlorite zone lie to the west of those from St. Pierre Lake in the biotite zone.

Magnetization ages

It is difficult to assign precise ages of magnetization to these four components because under conditions of elevated temperature followed by slow cooling the critical range of blocking temperatures is considerably lowered (Pullaiah *et al.* 1975) and also because they appear to have a chemical origin. Since all four components are secondary, there is no evidence that they have a thermoremanent magnetization (TRM) origin. Either a viscous partial TRM or more probably chemical remanent magnetization (CRM) (e.g., Dunlop 1985) is the preferred process of acquisition of magnetization. Component C is encountered in all the sampling sites (Table 1). This component is interpreted as representing the last metamorphic event. It is believed that metamorphism in the northernmost sector erased components older than component C. Locally, older remanence

TABLE 2. Site-mean directions of magnetization of Labrador Trough metabasalts and metagabbros after AF and thermal treatment

Site	N	(n)	*In situ* D (°)	*In situ* I (°)	*In situ* α_{95} (°)	*In situ* k_1	Tilt-corrected D (°)	Tilt-corrected I (°)	α_{95} (°)	k_2	k_2/k_1	FT	$H_{1/2}$ (mT)	T_{UB} (°C)	R
1980 GERIDO LAKE AREA															
Component A															
1	3	(6)	155	31	15.7	17	139	30							
2	4	(7)	140	27	5.5	28	128	18							1
3	4	(9)	139	23	8.4	19	127	11					20–35		
4	4	(9)	149	17	8.9	18	179	06						450–525	
5	2	(3)	105	56	—	—	231	13							
6	5	(10)	168	13	9.1	17	150	−18							
6	22	(44)	146	29	13.2	19	157	12	38.3	4	4.75	−			
Component B															
1	3	(3)	072	53	4.2	70	68	26							
2	1	(2)	061	49	—	—	61	19							
3	1	(1)	076	52	—	—	70	13					≤15	≤450	
4	2	(2)	034	53	—	—	92	−15							
5	1	(2)	052	63	—	—	78	−11							
6	3	(4)	070	29	16.7	19	61	−01							
6	11	(14)	62	51	12.5	30	72	05	17.2	16	1.88	±			
Component C															
1	4	(9)	275	16	15.7	34	284	37							
2	4	(8)	283	17	17.9	27	295	37							
3	3	(6)	264	19	27.3	21	020	−53					≤35	≤525	
4	2	(3)	281	05	—	—	354	−58							
5	3	(6)	273	26	25.3	24	310	−65							
6	3	(7)	270	23	28.9	17	029	−48							
6	12	(39)	275	18	8.2	67	327	−40	71.6	3	22.3	−			
Component D															
1	2	(2)	215	−18	—	—	216	07							
2	2	(2)	203	19	—	—	191	41							
4	2	(3)	223	23	—	—	182	−62					≤25	≤500	
6	3	(3)	214	10	21.3	35	103	−34							
4	9	(10)	214	09	23.4	16	178	−17	63	3	5.33	−			
1981 LAKE DOUBLET – LAKE RETTY – LAKE AULNEAU AREAS															
Component A															
M-1	3	(6)	138	38	15.2	66	190	−01					40–60	≤500–540	
M-5	2	(3)	130	48	—	—	135	31							
2	5	(9)	134	43			165	17	—	—					
Component B															
M-2	2	(3)	48	59	—	—	35	−05							
M-3	3	(5)	76	32	18.9	44	72	05					≤20	≤425	1
M-4	4	(7)	94	44	30.1	10	106	32							1
M-5	2	(4)	98	68	—	—	119	54							1
4	11	(19)	79	52	23.9	15.8	78	25	53.9	4	3.95	±			
Component C															
M-1	3	(7)	282	29	31.9	12.7	304	−49							
M-2	3	(12)	266	47	13.6	83	334	34					≤30	≤500	
M-3	5	(18)	277	13	2.1	30.3	286	32							2
M-4	3	(6)	286	02	37.0	12.2	293	−12							
M-5	5	(11)	280	−01	27.4	8.7	279	11							
5	19	(54)	279	11	20.4	15	298	05	40.8	4	3.75	−			
Component D															
M-1	3	(5)	184	18	8.4	216	161	−37							
M-2	2	(3)	185	20	—	—	175	35							
M-3	1	(2)	185	09	—	—	177	27						≤525	
M-4	4	(8)	190	28	22.3	18	187	−38							2
M-5	2	(3)	201	31	—	—	194	19							
5	12	(21)	190	22	14.1	44	179	02	37.1	5	8.8	−			

TABLE 2. (*Concluded*)

Site	N	(n)	*In situ* D (°)	*In situ* I (°)	*In situ* α_{95} (°)	*In situ* k_1	Tilt-corrected D (°)	Tilt-corrected I (°)	Tilt-corrected α_{95} (°)	Tilt-corrected k_2	k_2/k_1	FT	$H_{1/2}$ (mT)	T_{UB} (°C)	R
1982 LAKE PIO AND LAKE ST. PIERRE AREAS															
Component C															
M-11	5	(12)	288	21	18.2	18	348	43							1
M-12	4	(9)	277	−02	18.0	26	306	−25					10−30	400−425	
M-13	5	(10)	281	−01	21.2	13.8	296	−19							1
3	14	(31)	283	06	21.4	33.2	314	−02	62	4	8.3	−			
MEANS															
1980−1981 component A															
8*			143	33	14.4	15									
1980−1981 component B															
10*			69	51	10.3	23									
1980−1982 component C															
14*			278	15	7.7	28									
1980−1981 component D															
9*			200	16	13.2	16									

NOTES: N = number of samples; n = number of specimens; D = declination; I = inclination; k_1, k_2 = Fisher's (1953) precision parameter; α_{95} = cone of confidence at 95% level; $H_{1/2}$ = median destructive field; T_{UB} = unblocking temperature; R = number of reversed samples observed at a site; FT = folt test.
*Number of sites.

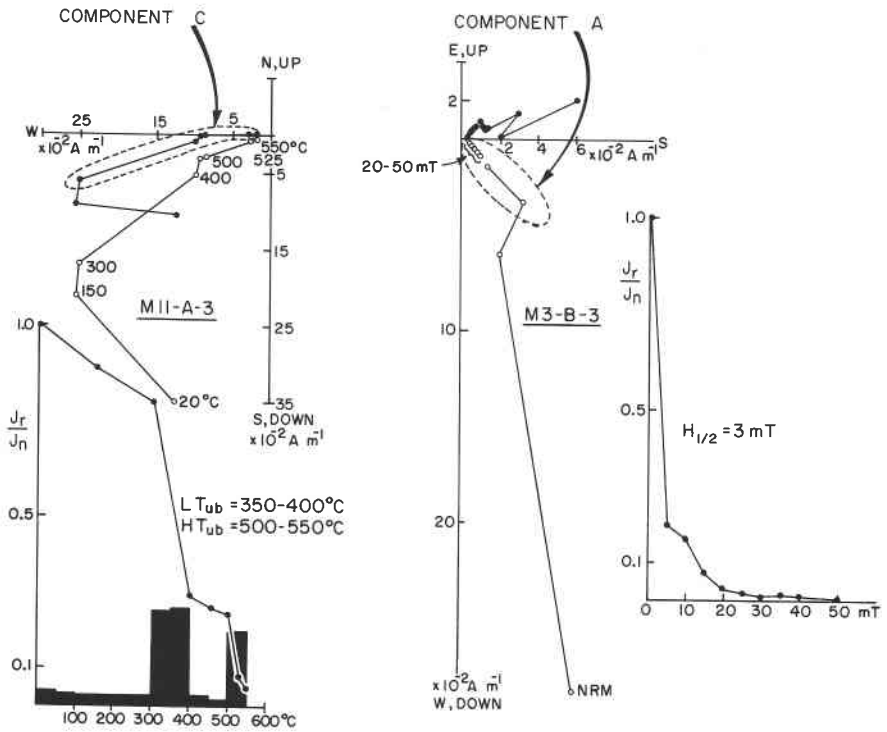


FIG. 4. Representative orthogonal thermal and AF demagnetization diagrams (Zijderveld 1967) of successive demagnetization vectors for two specimens as indicated (components A and C). Solid (open) circles represent projections on horizontal (vertical) planes. Numbers next to the open circles represent the demagnetization steps.

directions are preserved in the areas farther south, in rocks of lower metamorphic grade. Vector diagrams failed to establish a coherent sequence for the other three components. In addition, no mineralogical differences have been noted that might allow determination of the relative ages of components A, B, and C. These three components may all have a chemical remanent origin.

Comparison of the paleopole positions with the Middle Proterozoic APW of the Canadian Shield

Paleomagnetic mean poles for A, C, and D are given in Table 3 and plotted in Fig. 5. Pole positions of Proterozoic rocks in the Circum-Ungava Belt and adjacent areas of the Canadian Shield were presented by Schwarz and Freda (1983). Many paleomagnetic studies have been conducted in the

TABLE 3. Paleopole positions

Component	No. of sites	Longitude (°E)	Latitude (°)	δ_p (°)	δ_m (°)	Paleolatitude (°N)
		1980 Gerido Lake area				
A	6	323	11S	11.5	20.9	17
B	6	025	41N	11.4	16.9	32
C	6	201	10N	4.4	8.5	09
D	4	255	23S	11.9	23.6	03
		1981 Doublet Lake – Retty Lake – Aulneau Lake areas				
A	2	334	06S	—	—	25
B	4	012	32N	—	—	33
C	5	200	10N	10.5	20.7	06
D	5	283	23S	7.8	14.8	11
		1981 Pio Lake – St. Pierre Lake areas				
C	3	192	09N	10.6	21.0	03
		Mean (three areas)*				
A	8	327	09S	11	20	18
B	10	020	27N	—	—	32
C	14	199	11N	10	17	08
D	9	270	23S	10	18	08

NOTE: δ_p, δ_m = semi-axes of oval at 95% confidence level about the pole calculated from the mean direction.

*Calculated giving equal weight to data from each of the three areas.

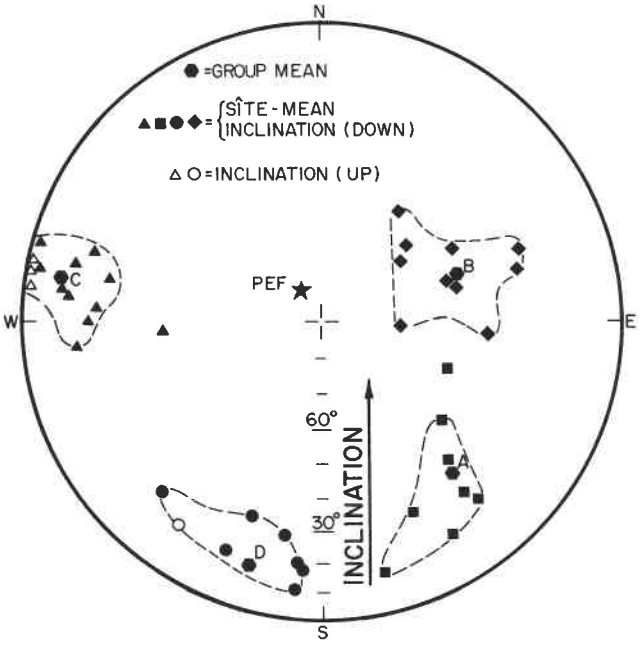


FIG. 5. Mean directions of magnetization per site and mean direction for each of the four components of magnetization (Table 2) plotted on a Schmidt net.

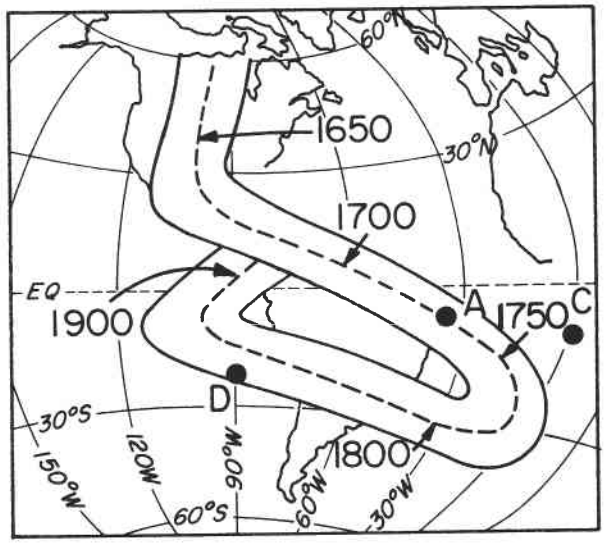


FIG. 6. Plot of poles for components A, C, and D on the suggested APW path for late Aphebian to Paleohelikian times (Irving and McGlynn 1981).

Labrador Trough (Seguin 1972, 1974, 1975*a*, 1976*a*–*d*, 1977*a*,*b*, 1978, 1979; Fahrig 1976; Park 1977; Schwarz and Freda 1983). Component D (200°, +16°) is comparable to Park's (1977) component A (203°, +16°) in the Seward Formation (mean of two sites) and Seguin's (1976*a*) thermally demagnetized component in the Nimish volcanics. The paleopole from the D component falls at 1850 Ma on the apparent polar wander (APW) path (Fig. 6) for the late Aphebian (Morel and Irving 1978; Irving and McGlynn 1981). Component A (143°, +33°) is comparable to Park's component A (one site; 154°, +21°) of the Menihek Formation. The paleopole of component A falls at 1740 Ma or slightly older on the APW path (Irving 1979; McGlynn and Irving 1981).

The paleopole of component C falls some 15° northeast of the large bend of the Coronation Loop (Irving and McGlynn 1981); it also compares with the Schwarz and Freda (1983) APW path. The age of magnetizations inferred from the APW path (Fig. 6) is ca. 1765 Ma. Assuming that the age of magnetization correlates with the age of a geological event, ca. 1765 Ma corresponds to the culmination of the Hudsonian Orogeny.

The paleopole position (20°E, 37°N) of component B (not plotted in Fig. 6) is comparable to the pole of a late component obtained in iron formations on the southern sector of the Labrador Trough (Seguin 1974, 1977*a*, 1979), which was then interpreted as an overprint related to the Elsonian Orogeny

(1350–1400 Ma ago). Unless related to the Elsonian orogenic event, component B remains unexplained. In summary, all the components of magnetization are overprints probably related to orogenic cycles. Very few reliable radiogenic ages for the Labrador Trough are available (Fryer 1972; Dressler 1975; Dimroth 1970). Most of these ages correspond to the Hudsonian Orogeny or are younger. A probable primary U/Pb zircon age of 2142^{+4}_{-2} Ma is available for acid volcanic rocks of the Mistamisk Formation of the central Labrador Trough (T. Krogh, *in* Clark 1984). There is thus a large spread between this primary age and that given by the paleopole positions. This supports our conclusion that the paleomagnetic age is of orogenic origin.

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